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Paleomagnetism of the Jurassic Transantarctic Mountains revisited — Evidence for large dispersion of apparent polar wander within less than 3 Myr

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ABSTRACT

The Transantarctic mountains predominantly consist of Jurassic continental flood basalts (Kirkpatrick) and sills (Ferrar) emplaced during the earliest phase of the break-up of Pangea. Published ages, based on a variety of geochronological methods all agree rather well and cluster around 180 Ma suggesting emplacement during a rather short time interval of not more than 3 Myr. Paleomagnetic studies, mostly carried out between the 60s and 90s of the last century, however, yield two well defined but significantly different groups of paleomagnetic pole positions plotting either in intermediate latitudes (A) or at latitudes exceeding 60°S (B). Pole positions belonging to group A are generally interpreted to be of primary origin whereas the significance of group B poles remains unclear. Here we report new data from the Kirkpatrick basalts of Northern Victoria Land, Antarctica, where 157 oriented paleomagnetic samples were taken, covering about 800 m of stratigraphy and 23 volcanic flows in Gair Mesa (73.4666°S; 162.8666°E). After removal of a steep magnetic overprint with rather low coercivities straight linear segments of exclusively normal polarity trending toward the origin of projection are identified in 151 samples from 22 sites. Maximum unblocking temperatures do not exceed 580 °C *and* maximum coercivities not 60 mT. The resulting mean virtual geomagnetic pole (VGP) for the Gair Mesa plots at 66.4°S; 227.7°E (α_{95} of 6.2°, k = 25.7) comparable to group B poles. Reflecting light microscopy reveals that the magnetic inventory of all samples analyzed is dominated by magnetite showing shrinking cracks and broad ilmenite lamellae, the latter being diagnostic for high temperature oxidation. Analysis of the distribution of the 22 pole positions obtained here shows that secular variation has successfully been averaged (S_b = 15.9°) suggesting that the rocks studied here have recorded the time averaged geomagnetic field during early mid Jurassic times. Based on these new results we postulate that both clusters A and B of pole positions reflect primary magnetizations and that the Kirkpatrick basalts of Gair Mesa have been emplaced in a rather short normal polarity time interval of 3 Myr. If our reasoning is correct, the apparent polar wander path for the latest early Jurassic might be more complex than previously thought. Whether this complexity is evidence for massif True Polar Wander remains speculative.

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1. Introduction

The exact shape of the apparent polar wander path (APWP) for East Antarctica during Jurassic times is still a matter of some debate and complicated by the presence of two groups of pole positions which plot either at latitudes of around 50°S (group A) or around 70°S (group B). In their study of Ferrar dolerites of Jurassic age in the Prince Albert Mountains Lanza and Zanella (1993) recognize pole positions belonging to either one of these two groups and interpret *intermediate latitude* pole positions (group A) to be representative for the *true* Jurassic paleopole for Antarctica, whereas the *high latitude* cluster (group B) is thought to reflect magnetizations also acquired during the Jurassic but while the Earth magnetic field went through a transitional phase e.g. a change in polarity. Alternatively, it has been argued repeatedly that the poles of group B are of secondary origin (Cherry & Noltimier, 1981; Delisle & Fromm, 1984, 1989; Rolf & Henjes-Kunst, 2003) and are linked to a Cretaceous thermal event (Molzahn et al., 1999). The latter being associated with the down faulting of the Rennick Graben (Luyendyk et al., 1996), separating East and West Antarctica in post-Early Cretaceous time (Grindley & Oliver, 1983; Bentley, 1991) or an early phase of uplift of the Transantarctic mountains triggering a change in the groundwater and heat flow regime (Fleming et al., 1992). To some extent, these interpretations are based on the resemblance of the group B poles to the mean Cretaceous pole position for West Antarctica published by DiVenere et al. (1994, 1995). Yet, a third explanation has been brought forward by McIntosh et al. (1986) invoking incomplete averaging of secular variation as cause for the

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high latitude pole positions. In any case, which of these various alternatives will prove to be valid will have severe impact on either our understanding of the fine structure of the Earth's Magnetic Field and the speed of emplacement of the Karoo–Ferrar Large Igneous Province.

2. Geological setting and age of the Kirkpatrick basalt

The geology of Northern Victoria Land (NVL) has been described in detail (Stump, 1986; Faure & Mensing, 2010, and references therein) and is repeated only briefly here. The basement rocks of NVL, first described by McLeod and Gregory (1967) and followed by Tessensohn et al. (1981) consist of paragneisses of the Wilson Group (Cambrian), the metavolcanic and metasedimentary rocks of the Bowers Group (Late Camrian to Ordovician), and the greywackes and argillites of the Robertson Bay Group (Late Precambrian (Stump, 1995), and references therein). Northern Victoria Land also contains comparatively small remnants of the Beacon Supergroup of Paleozoic age and volcanic rocks of the Ferrar supergroup of Jurassic age. The latter are very prominent in the Mesa Range at the head of the Rennick Glacier.

The youngest igneous rocks in Northern Victoria Land are the alkali-rich basalt flows of the McMurdo Volcanic Group of Neogene age which crop out primarily along the coast of the Ross Sea.

The lithology of the Ferrar Supergroup consists of sills of the Ferrar Dolerite, intruding the Paleozoic and early Mesozoic sediments of the Beacon Group, of flows of the Kirkpatrick basalt, and of phreatomagmatic volcanic breccias that underlie the basaltic plateaus. All these rocks formed in a short interval of time during the latest Middle Jurassic in an extensional setting. The Kirkpatrick tholeiitic basalts have first been described in Southern Victoria Land by Gunn (1962) cited by Elliot (1972). They are interpreted to be co-magmatic and coeval with the Ferrar dolerite sills (Encarnacion et al., 1996). The basalt magmas originated in the subcrustal mantle and pierced through the basement of the Transantarctic Mountains (TAM) and the overlying Beacon Supergroup along deep crustal rifts caused by the breakup of Gondwana. Since the pioneering work of Gunn (1962), these basalts (now known as the Kirkpatrick basalt), have been identified almost everywhere along the Transantarctic mountains stretching from the Ross to the Wedell Sea and broadly following the outline of the Rennick Graben over a length of about 3500 km which marks the boundary between East and West Antarctica (Fig. 1). In northern Victoria Land the Kirkpatrick basalts reach an exposed thickness of about 800 m (Heimann et al., 1994) made up of at least 40 individual flows which so far have been identified (Elliot et al., 1986). No indication of post depositional tilt has been observed neither at the sampling location nor been described in the literature for other exposures of Kirkpatrick basalts in the Transantarctic Mountains.

Published ages for the Ferrar sills and Kirkpatrick basalts range from 176.6 Ma based on $^{40}Ar/^{39}Ar$ dating over 178.5 Ma (Rb–Sr) to 183.6 Ma (U–Pb Fleming et al., 1997). Age variations in the $^{40}Ar/^{39}Ar$ data set are believed to be systematic and to either reflect problems in calibration (Fleming et al., 1997) or Ar loss (Encarnacion et al., 1996). However, a thorough re-analysis of the existing $^{40}Ar/^{39}Ar$ data and the recalculation to a common standard (Renne et al., 1994) yields a mean age for Ferrar volcanic rocks of 180.3 ± 2.2 Ma (Riley & Knight, 2001; Riley et al., 2009). This age is based on single mineral analysis and includes U–Pb data (Fleming et al., 1997) but does not include K/Ar nor Rb/Sr ages published earlier and now interpreted to be obsolete (Riley & Knight, 2001). The 180 Ma age is comparable to age dates published for the Karoo province of southern Africa, yielding a mean age of 183.0 ± 1.0 Ma (Duncan et al., 1997) suggesting a duration for the volcanic activity of the Karroo/Ferrar large igneous province (LIP) of about 3–4 Myr (Riley & Knight, 2001) overall. Within this framework emplacement of the Kirkpartrick volcanic rocks is thought to have occurred during one rather short burst in the order of one million years or even less (Fleming et al., 1997).

3. Field procedures and laboratory methods

About 700 m of Kirkpatrick basalts are exposed at Gair Mesa situated at 73.4667°*S*; 162.8667°*E* (see also Fig. 1), a total of 157 cores were taken from 23 individual lava flows at Gair Mesa whereby each flow sampled was statistically treated as one site (Table 1). All samples have been cored in the field using a gasoline driven portable rock drill and oriented using both a sun compass and a southern hemisphere magnetic compass. Sun compass orientation data was transferred into the magnetic reference system using the 'sundec.py' algorithm of Tauxe (2010). Samples were subsequently studied in the paleomagnetic laboratory of the Ludwig-Maximilians-University, Munich. After cutting into standard ~10 *cc* paleomagnetic cylinders, pilot samples were demagnetized both in alternating fields (AF) and by stepwise heating (TH) and subsequently measured using 2G Enterprises cryogenic SQUID magnetometers housed in a magnetically shielded room. AF demagnetization was carried out using the home-made automated system at the University of Munich (Wack & Gilder, 2012) with peak fields of 90 mT. Demagnetization results were plotted on orthogonal vector diagrams (Zijderveld, 1967) and analyzed using the least square method (Kirschvink, 1980) on linear portions of the demagnetization paths defined by at least four consecutive demagnetization steps. All linear fits were anchored to the origin of the orthogonal projection and mean angular deviation (MAD) values of the computed linear fits are generally below 10°, mostly below 5°.

For the determination of either hysteresis parameters and of Curie temperatures (T_c) a Variable Field Translation Balance (VFTB, Krása et al., 2007) was used.

4. Results

4.1. Rock magnetism

As to be expected from the directional characteristics the rock magnetic properties of the material studied are very uniform. Thermomagnetic experiments in weak fields of 30 mT yield Curie temperatures (T_c) between 550 °C and 570 °C (Fig. 2a). The extremely high degree of reversibility indicates that no mineral alteration occurs during the heating cycle (Fig.2a). Acquisition of Isothermal Remanent Magnetization (IRM) shows rapid saturation in all 24 samples studied at around 200 mT (Fig. 2b) suggesting the presence of magnetite and showing no indication for the presence of any mineral with higher coercivity. Hysteresis parameters determined on a Variable Field Translation Balance (VFTB, (Krása et al., 2007)) are plotted in Fig. 2c and d. The distribution of hysteresis data on the day plot, even though it is a rather crude approach, plots along a proposed mixing curve (Dunlop, 2002, Fig. 2), which suggests the presence of a wide range of grain sizes.

In order to obtain additional information on the magnetomineralogy of the material, at least one specimen per site was polished and further analyzed microscopically under reflecting light with and without magnetic colloids. In all samples analyzed, low temperature oxidation is prevailing and can be identified by a slight change in color from redish brown to a rather more grayish shade (Fig. 3a). In addition, irregular curved cracks are identified and interpreted as shrinkage cracks (Petersen et al., 1979). Broad ilmenite lamellae, observed in almost all samples and especially pronounced when applying magnetic colloid to the polished surface points toward initial high temperature oxidation and the exsolution of titano-magnetite into magnetite and ilmenite (Fig. 3b). Being typical for ocean floor basalts low temperature oxidation does not affect the original directional Natural Remanent Magnetization (NRM) acquired by titanomagnetites during initially cooling below T_c (Matzka et al., 2003). Based on these observations we postulate that a primary magnetization has survived in samples from Gair Mesa. In addition, reflecting light microscopy on samples from the Litell Group (entry 18 in Table 2, Rolf & Henjes-Kunst, 2003)

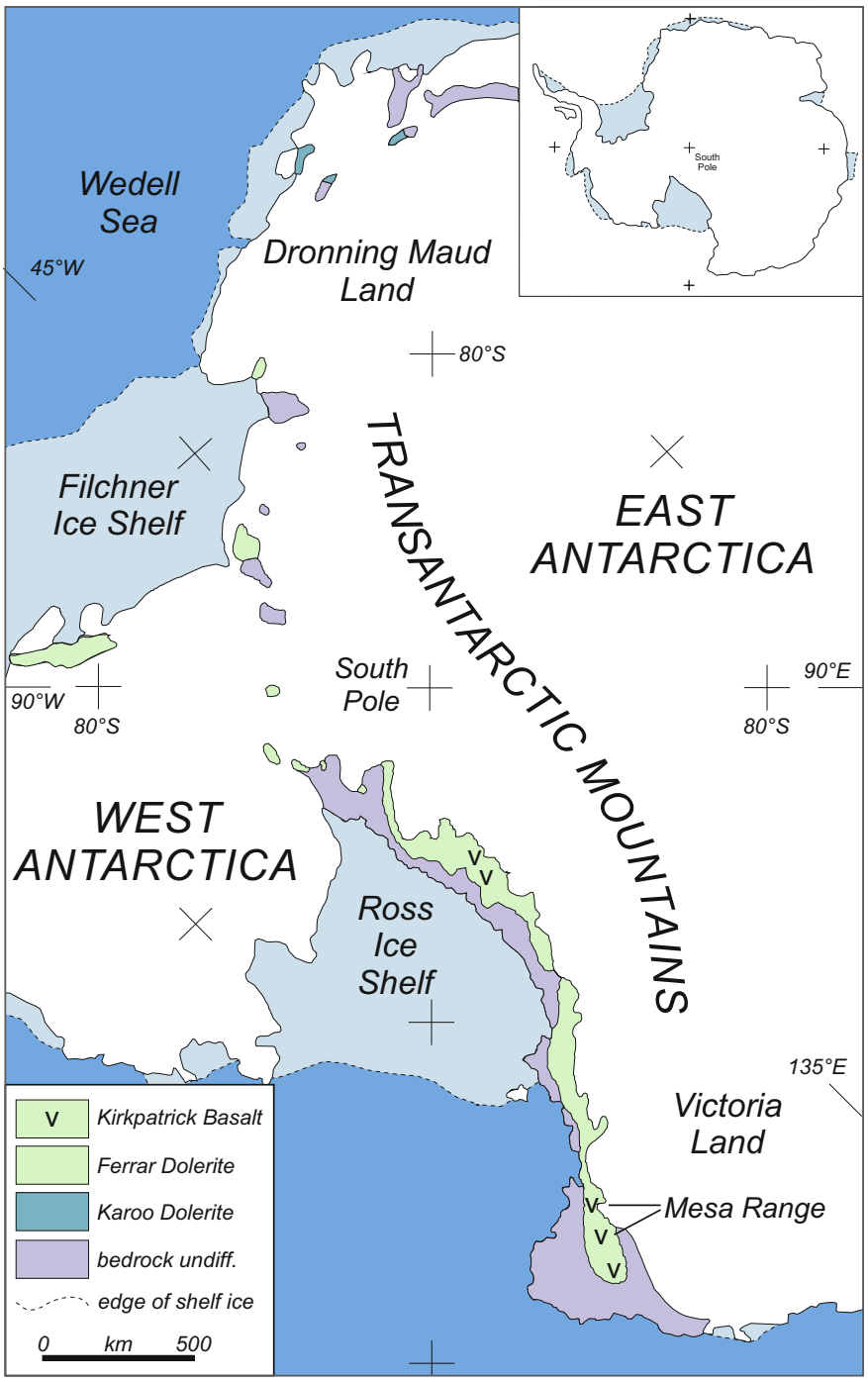


Fig. 1. Outcrops of igneous rocks of Jurassic age in Antarctica (after Fleming et al., 1997). The sampling area (Mesa Range) is marked.

reveals all the indications for high temperature oxidation and a complete lack of secondary low temperature alteration (Fig. 3c).

4.2. Paleomagnetism

4.2.1. Demagnetization results

Almost all samples studied paleomagnetically show rather simple directional behavior independent of the demagnetization method applied (Fig. 4). After removal of a viscous overprint generally below 150 °C a monodirectional (southwesterly and upward) component of magnetization of exclusively normal polarity, trending straight into the origin of the orthogonal projection (Zijderveld, 1967) can be identified which reaches maximum unblocking temperatures around 560 to 570°C (Fig. 4a and b). Similar behavior can be observed during demagnetization in alternating fields (AF). Again, a viscous component of magnetization is being removed in fields of up to 20mT. Above this peak field field, again, a southwesterly and upward pointing component of magnetization is identified, reaching maximum coercive forces of 90 mT (Fig. 4c to f). Both demagnetization methods yield virtually indistinguishable characteristic remanent magnetizations (ChRM) and therefore the remainder of the collection was subjected to demagnetization in alternating fields. The resulting mean direction, based on characteristic sample directions plots at Dec: 289.0°, Inc: −78.1° with an α_{95} of 1.8° and a k-value of 39.6 based on 151 samples (Fig. 5a) whereas the resulting overall site mean direction based on 22 sites plots at Dec: 287.8° Inc: −80.1° with an α_{95} of 3.9° (Fig. 5b) with a

Table 1
Site mean directions for Kirkpatrick basalts, Gair Mesa.

Site	N	GLAT [°S]	GLONG [°E]	m	D [°]	I [°]	k	α_{95} [°]	PLAT [°S]	PLONG [°E]
MR1	8/8	73.3452	162.9165	~10	292.1	−84.5	21.0	12.4	74.9	202.8
MR2	11/12	73.3397	162.7953	143	313.5	−72.6	102.6	4.5	66.4	268.4
MR3	10/10	73.3451	162.9182	380	300.1	−68.9	87.2	5.2	57.8	260.1
MR4	6/6	73.3452	162.9178	390	297.3	−61.7	131.9	5.9	48.4	263.9
MR5	15/17	73.3455	162.9207	470	300.0	−76.8	157.6	3.1	68.3	244.5
MR6	12/12	73.3713	162.8653	~10	238.7	−81.2	52.7	6.0	60.6	193.8
MR7	7/7	73.3713	162.8650	100	256.2	−78.4	30.3	11.1	59.3	209.3
MR8	6/6	73.3713	162.8650	100	261.0	−78.9	602.1	2.7	61.2	211.1
MR9	5/5	73.3710	162.8639	~10	271.4	−67.4	62.8	9.7	47.8	235.0
MR10	n/r	73.3710	162.8639	~10						
MR11	5/5	73.3199	162.8975	700	177.5	−79.9	23.3	16.2	53.7	161.5
MR12	6/6	73.3261	162.9131	650	241.9	−74.7	17.9	16.3	50.9	205.0
MR13	5/5	73.3276	162.7491	380	309.7	−86.6	585.6	3.2	76.6	185.6
MR14	7/7	73.3276	162.7491	380	283.2	−79.7	336.3	3.3	67.4	222.3
MR15	7/7	73.3271	162.7499	422	271.1	−82.9	187.9	4.4	68.5	204.2
MR16	5/5	73.3275	162.7512	475	263.9	79.3	342.0	4.2	62.3	211.9
MR17	2/5	73.3452	162.9179	380	255.2	−71.5			49.0	218.1
MR18	5/5	73.3452	162.9179	380	294.3	−81.7	106.9	7.4	72.3	220.2
MR19	6/6	73.3459	162.9195	402	304.7	−74.6	163.1	5.3	66.6	254.7
MR20	6/6	73.3460	162.9207	400	309.9	−70.0	176.3	5.1	62.0	268.9
MR21	4/4	73.3462	162.9207	435	306.2	−75.5	346.3	4.9	68.2	253.8
MR22	6/6	73.3460	162.9216	435	323.5	−75.0	66.1	8.3	72.4	274.5
MR23	7/7	73.3464	162.9220	475	314.2	−79.6	24.9	11.2	75.5	244.2
Mean	22/23				285.4	−78.3	82.	3.4		
					Mean	Pole:	$k = 25.7$	$A_{95} = 6.2°$	66.4°S	227.7°E

Site, sampling sites; *N*, number of samples used for calculating site mean directions/number of measured samples (n/r, not recorded); GLAT and GLONG, geographic coordinates of sampling site; m gives the approximate height of sampling site above top of ice; *D*, Declination; *I*, Inclination in geographic coordinates; *k*, Fisher precision parameter (Fisher, 1953); α_{95}, Fisher circle of 95% confidence; PLAT and PLONG, position of according paleopole.

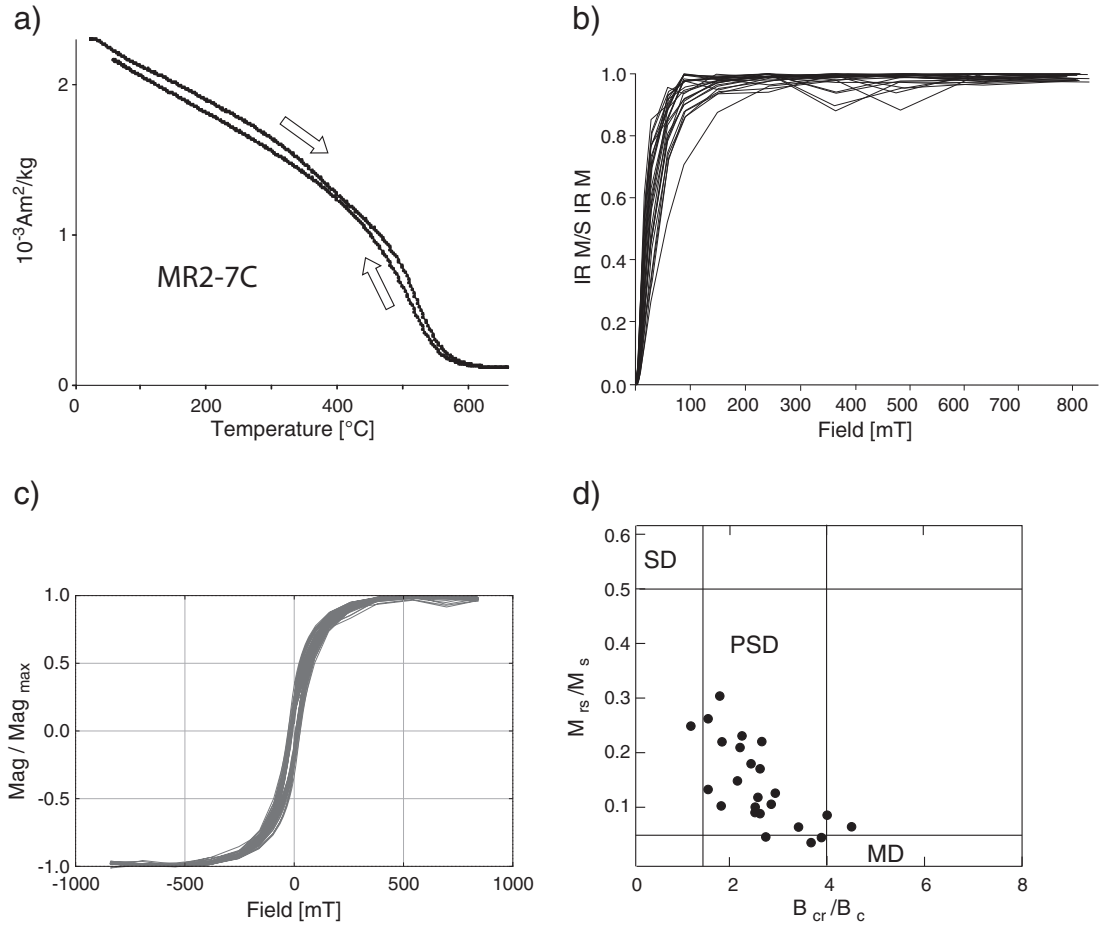


Fig. 2. (a) Thermomagnetic curve of a representative sample showing heating and cooling cycles. (b) Acquisition of Isothermal Remanent Magnetization for 24 samples (normalized). (c) Hysteresis data for 24 samples (normalized). (d) Summary of hysteresis data (Day et al., 1977).

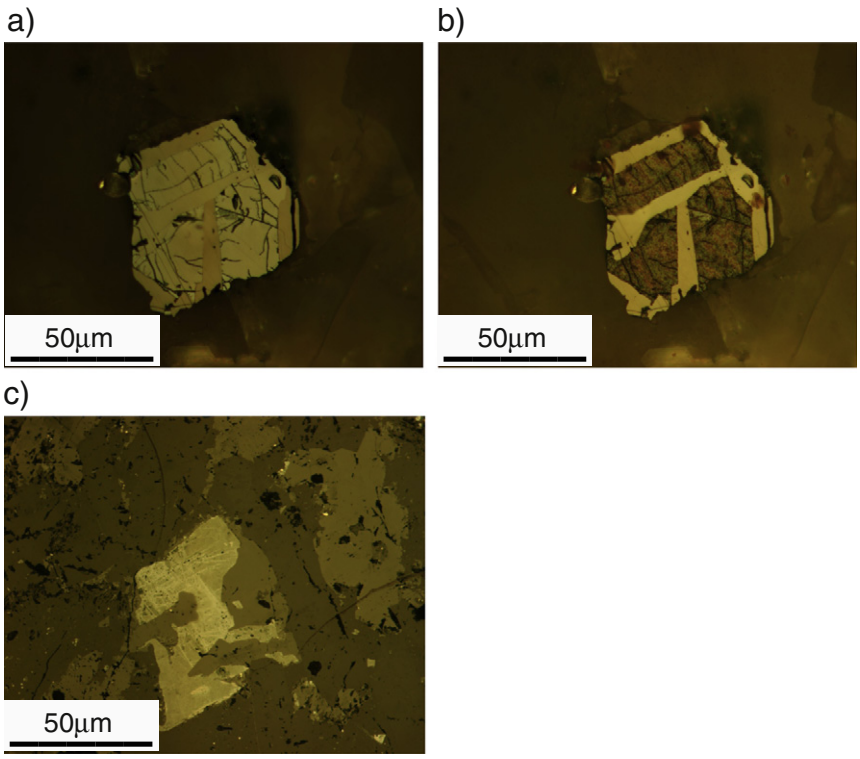


Fig. 3. Reflected light microscopy of sample MR3–5. Without (a) and with magnetic colloid (b). Note the broad ilmenite lamellae and the curved shrinkage cracks. Also shown is one image from a sample from the Litell Group (entry 18 in Table 2, (Delisle & Fromm, 1984)) showing no evidence for low temperature oxidation.

resulting paleopole position at 66.4°S, 227.7°E and an A_{95} of 7.0° (Tables 1 and 2). The 22 site mean directions yield a within-site corrected secular variation value (S_b, Biggin et al., 2008) of 15.9°.

4.2.2. A new paleopole position for the Kirkpatrick basalt

In order to compare the result from Mesa Range to reference data, the recently published apparent polar wander path for Pangea (Kent & Irving, 2010) was transferred into East Antarctica coordinates. The VGP obtained in this study plots rather close, yet displaced to the south by some ten degrees to the cluster of poles ranging in age from 160 to 190 Ma and defining the early to mid Jurassic standstill (Kent and Irving, 2010). No similarity to younger pole positions (e.g. Cretaceous) can be observed (Fig. 6).

5. Comparison of the Mesa Range VGP to published Data

Combining the resulting paleopole position obtained in this study with published data from the Ferrar and Karoo Supergroups of Eastern Antarctica yields a total of 28 entries listed in Table 2 and plotted on Fig. 7. No quality criteria such as proposed by Van der Voo (1993) have been applied and we are fully aware that quite a substantial number of entries will fail if more stringent selection criteria were to be applied. Many of the oldest results from the late fifties to late sixties have not been subjected to modern demagnetization techniques nor have the results been documented in such a way that they conform to modern standards. A further complexity arises from the fact that a significant number of entries in Table 2 are based on 1 or 2 sites only or that only the number of samples are reported without any reference to the sampling strategy applied. However, and despite some of these shortcomings, it is remarkable that almost all data listed in Table 2 plot along a well defined swath along 220°E and bracketed by 40°S and 79°S (Table 2, Fig. 7). Common to almost all pole positions listed in Table 2, however, are the relatively large values for k associated with relatively small values for α_{95} (Fisher, 1953). This raises the suspicion that secular variation has not always successfully been averaged out. In order to test individual entries of the compiled data set for secular variation, we applied the algorithm published by Deenen et al. (2011). Where necessary, the Fisherian parameters (k and α_{95}, Fisher, 1953) for individual results, taken from either the original publication or from the Global Palaeomagnetic Data Base (Global Paleomagnetic Database, GPMDB (McElhinny & Lock, 1996) version 4.6 updated February 2005) were converted into K following Cox (1970) and into A_{95}, accordingly. The latter using the approximation $A_{95} = \sqrt{\frac{140}{KN}}$ suggested by Irving (1964), where in this case N defines the number of sites used to calculate the VGP or if not available the number of samples. Both parameters (K and A_{95}) are deemed to more realistically describing the statistical properties of the distribution of pole positions. Only 8 out of 29 pole positions have associated values of A_{95} smaller than to be excepted for a given number of sites if secular variation was successfully averaged (Deenen et al., 2011). We note that pole positions failing the test for secular variation include members of both groups A and B. None of the VGPs listed in Table 2 has A_{95} exceeding the upper confidence limit of Deenen et al. (2011). The north south elongated uniform distribution of VGPs cannot be simplified, however, by either applying more stringent quality criteria or omitting results which might reflect secular variation or are based on insufficient numbers of samples (Fig. 7).

6. Discussion

6.1. Early Cretaceous remagnetization?

A secondary origin of paleopole positions of group B affinities has been postulated repeatedly (Delisle & Fromm, 1989; Faure & Mensing, 1993, 2010) and linked to the onset of separation of East and West Antarctica (Faure & Mensing, 1993) along the Rennick Graben. This was presumably supported by K–Ar whole rock studies yielding minimum ages of 103 ± 2 Ma for altered basalts (Faure & Mensing, 1993). Further support for the remagnetization concept arises from Ar-Ar and

Table 2
Paleopole positions for Ferrar and Karoo volcanic rocks from the Transantarctic Mountains and Dronning Maud Land, East Antarctica and Tasmania.

#	Formation	Location		Pole position					Reference
		Lat.	Long.	Lat.	Long.			A_{95}	
		[°S]	[°E]	[°S]	[°E]	N	K	[°]	
Transantarctic mountains									
Group A Poles									
1	David Glacier	75.87	158.50	55.0	222.0	25	48.5	4.0[b]	Cherry and Noltimier (1981)
2	Queen Alexandra Range	84.33	164.00	50.2	226.1	33[a]	58.0	5.1	Hoffman et al. (1984)
3	Queen Alexandra Range	84.20	165.00	54.2	220.2	26	39.1[a]	8.5	Ostrander (1971)
4	Wright and Victoria Valley	78.00	162.00	45.0	220.0	46	25.9	4.1	Bull et al. (1962)
5	Wright Valley	77.50	161.50	45.3	208.0	26[a]	53.9	3.7	Funaki (1984)
6	Ferrar Glacier	78.00	161.00	58.0	218.0	5	16.5	15.4	Turnbull (1959)
7	Beardmore glacier	84.00	195.00	59.0	221.0	9	5.9	19.2	Briden and Oliver (1963)
8	Dufek intrusion	82.60	307.50	60.0	223.0	91[a]	11.8	6.6[b]	Burmester and Sheriff (1980)
9	Theron Mts.	80.50	335.00	54.0	231.0	8	29.5	18.9	Blundell and Stephenson (1959)
10	Ohio Range, Mt. Schopf	84.75	246.40	58.0	231.0	8[a]	19.3	11.2[b]	Kellogg and Reynolds (1978)
11	Mt. Cerberus	77.43	161.88	57.8	224.3	9[a]	80.8	7.8[b]	Kellogg and Reynolds (1978)
12	Allan Hills A	76.70	159.70	42.2	226.2	14	20.6	8.2[b]	Funaki (1984)
13	Carapace Nunatak	76.88	159.40	40.9	196.6	4[a]	31.2	12.5	Funaki (1984)
14	Northern Prince Albert A	75.30	161.00	47.8	226.0	15	50.2	5.1	Lanza and Zanella (1993)
15	Sequence Hills	73.05	161.25	57.7	220.0	8[a]	6.4	19.5	Delisle and Fromm (1989)
Group B poles									
16	DeGoes Ridge and Johnstone Glacier	71.74	161.90	74.7	121.7	6	17.8	16.6	Delisle and Fromm (1984)
17	Morozumi and Lanterman	71.00	162.50	70.8	225.9	7	11.0	15.9	Delisle and Fromm (1984)
18	Litell	71.75	162.50	78.1	219.5	4	39.3	8.4	Delisle and Fromm (1984)
19	Kirkpatrick Basalt	75.25	162.00	64.0	210.1	15	278.0	4.1	McIntosh et al. (1986)
20	Allan Hills B	76.70	159.70	62.5	229.9	5[a]	146.1	5.2[b]	Funaki (1984)
21	Mt. Fleming	77.55	160.10	68.9	220.5	15[a]	412.0	1.8[b]	Funaki (1984)
22	Northern Prince Albert B	75.30	161.00	79.0	181.0	7	97.3	6.1	Lanza and Zanella (1993)
23	Ferrar Group	71.75	162.00	72.0	236.0	8	28.0	9.3	Rolf and Henjes-Kunst (2003)
24	Roberts Butte	72.65	160.13	76.9	219.1	8[a]	12.5	14.0	Delisle and Fromm (1989)
25	Gair Mesa	73.33	162.90	66.4	222.7	22	22.9	7.0	This study
Dronning Maud Land									
26	Vestfjella	80.00	331.75	41.8	226.5	32	33.5	4.3	Løvlie (1979)
27	Petermannketten P2	71.36	12.60	59.0	237.0	28[a]	120.0	2.5[b]	Rolf and Henjes-Kunst (2005)
28	Petermannketten P3	71.41	12.84	59.0	235.0	40[a]	494.0	1.0[b]	Rolf and Henjes-Kunst (2005)
Tasmania									
TD	Tasmanian dolerites			62.6	219.1	34	13.2	6.6	Schmidt (1976)

Unless given in the original publication K has been recalculated after Cox (1970) and A_{95} after Irving (1964). The pole position for the Tasmanian dolerite is given in East Antarctic coordinates Williams et al. (2011). Labels refer to Fig. 7.
Pole positions where secular variations has not been averaged (entries 1,8,10,11,12,20,21,27,28) or based on less then than 16 samples (entries 8,10, 11,13,20,24) are shaded in light gray in Fig. 7.
[a] Number of samples.
[b] A_{95} below the lower reliability envelope of Deenen et al. (2011) suggesting that secular variation might not have been averaged.

K–Ar studies of Ferrar apophyllites (Fleming et al., 1999; Molzahn et al., 1999) dating thermal events postdating the emplacement of the Ferrar volcanics by up to 80 Ma. However, since formation of apophyllite is related to groundwater flow and uplift at rather low temperatures, it is more than unlikely that a link between zeolite (e.g. apophyllite) formation and remagnetization can be established. We also point out that remagnetization cannot be reconciled with the results of our microscopical examination of the oxidation state of magnetite in samples from Mesa Range. These arguments are supported by the findings of (Antonini et al., 1997) who does not recognize any post emplacement alteration resulting in a reset of the Rb/Sr system in Kirkpatrick basalts. However, as pointed out by Antonini et al. (1997), volcanic rocks from the Prince Albert Mts. (Lanza & Zanella, 1993) yielding intermediate latitude VGPs (group A) have different chemical signatures when compared to rocks yielding high VGP latitudes (group B), suggesting different styles of magmatic differentiation. Based on these arguments, an early Cretaceous remagnetization event resulting in group B type pole positions is not tenable anymore. As pointed out earlier (McIntosh et al., 1986) paleomagnetic data from Tasmania lends additional support to this line of argument (McIntosh et al., 1986). The Tasman dolerite which is geochemically and geochronologically indistinguishable from the Ferrar province, studied by Irving and Parry (1963), Schmidt (1976) carries a primary magnetization, supported by a positive conglomerate as well as baked contact test (Irving & Parry, 1963), and yields a paleopole position at 62.6°S and 219.1°E in East Antarctic coordinates (Williams et al., 2011) falling into group B.

6.2. Secular variation or transitional directions?

Alternatively, it has been speculated that results of group B are affected by secular variation and therefore do not represent the time averaged dipole field (McIntosh et al., 1986) or that group B pole positions mark the onset of a polarity transition (Lanza & Zanella, 1993). However, whereas secular variation seems to be rather unlikely to have affected the data set, the onset of a polarity transition might be feasible. Yet, the obviously well organized pattern of group B paleopole positions does not fit typical reversal patterns which are more chaotic and marked by significant jumps (see for example Singer et al., 2005; Valet et al., 2012).

Also, the secular variation value (S_b) of 15.9° observed in the Gair Mesa basalts is in general agreement with published data for the Jurassic (Biggin et al., 2008). A starting polarity transition would probably cause an elevated secular variation value. An additional ambiguity arises from the fact that no absolute paleointensity data is available for any of the rocks listed in Table 2 therefore not allowing to test for reduced intensities of the Earth magnetic field during emplacement of rocks carrying group B magnetizations.

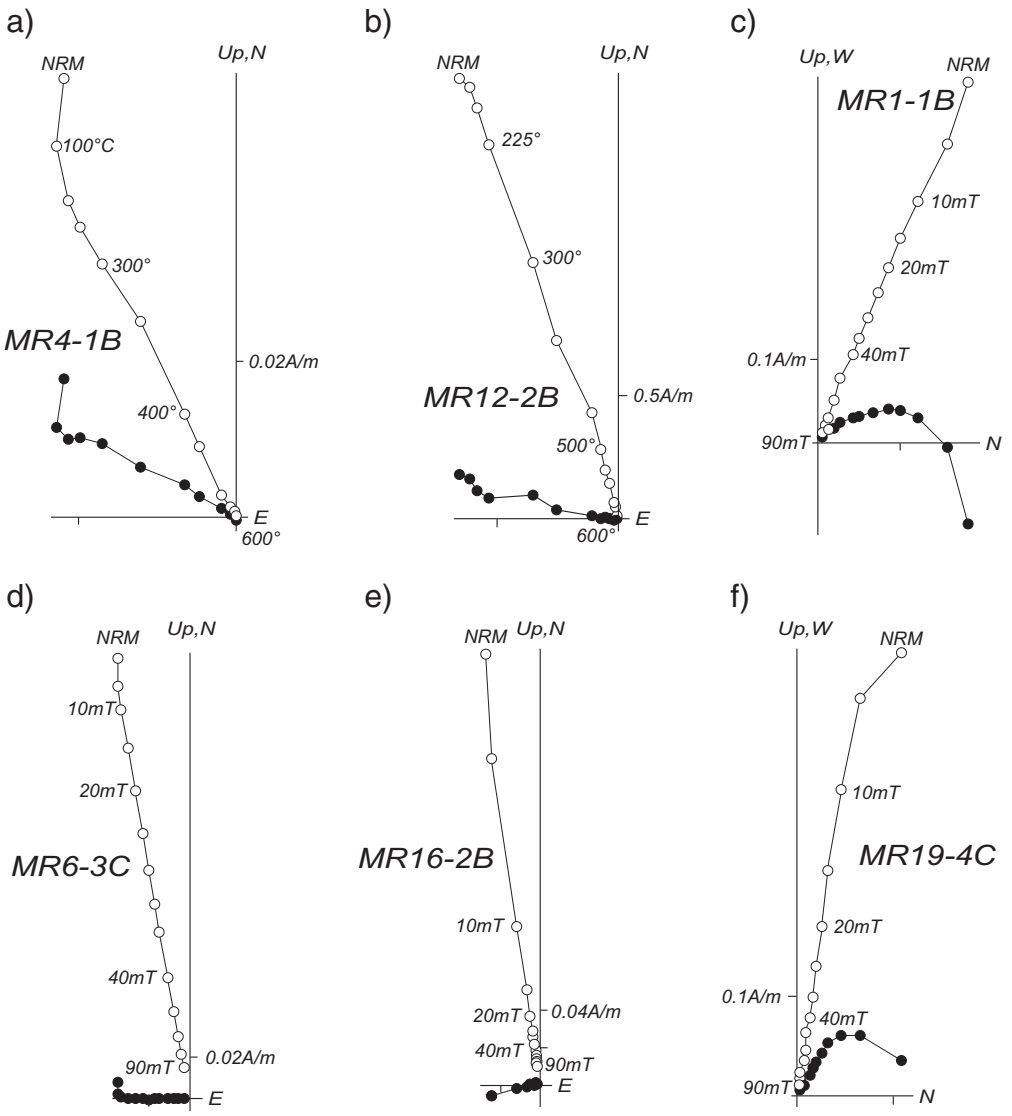


Fig. 4. Orthogonal projection of demagnetization data (Zijderveld, 1967). Solid (open) circles correspond to projection onto the horizontal (vertical) planes. a) and b) show the results of thermal demagnetization experiments, c) to f) of demagnetization in alternating fields. Selected steps in milli Tesla and °C are indicated.

6.3. Significance of Magnetic Polarities and implications for the duration of emplacement of Ferrar and Kirkpatrick rocks

With the exception of the results from the Dufek intrusion (Beck, 1972), Mt. Theron (Blundell & Stephenson, 1959) and Vestfjella (Løvlie, 1979), entries 8, 9, 26 in Table 2 all directional data reported from volcanic rocks from Eastern Antarctica are of normal (upward pointing inclination) polarity. The significance of the mixed polarity data, however, is still unclear. Unfortunately, no detailed information about the magnetic properties of the Dufek samples has been given in the original paper (Beck, 1972) and their directional behavior during demagnetization cannot be judged. However, the reversal test is inconclusive (GPMDB) and it is not clear whether the inverse polarity directions are actually representing a reversed state of the Earth Magnetic Field or do reflect a self reversing mechanism caused by low-temperature mineralogical alteration (Beck, 1972). Mixed polarities identified in the Theron Mts. (Blundell & Stephenson, 1959) are based on the directions of the NRM of eight paleomagnetic samples which have not been subjected to any demagnetization experiments. Nevertheless the reversal test (Tauxe, 1998) is positive and the distribution of normal (NE and up) and reversed data (SW and down) bears a

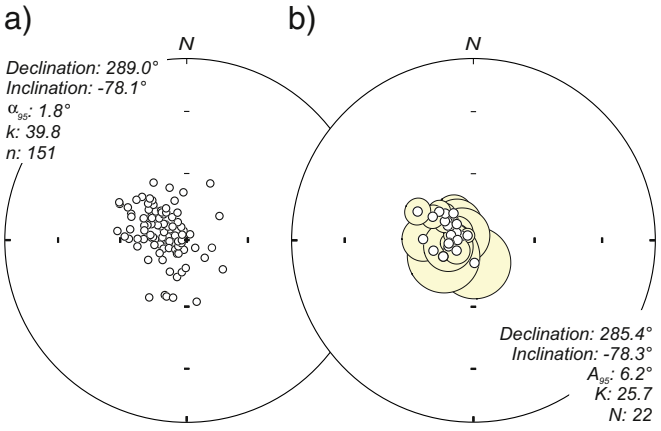


Fig. 5. Directions of the characteristic remanent magnetization for 151 individual samples (a) and resulting site mean directions for 22 sites. Open circles are projections into the upper hemisphere. Also, the corresponding overall mean direction is given.

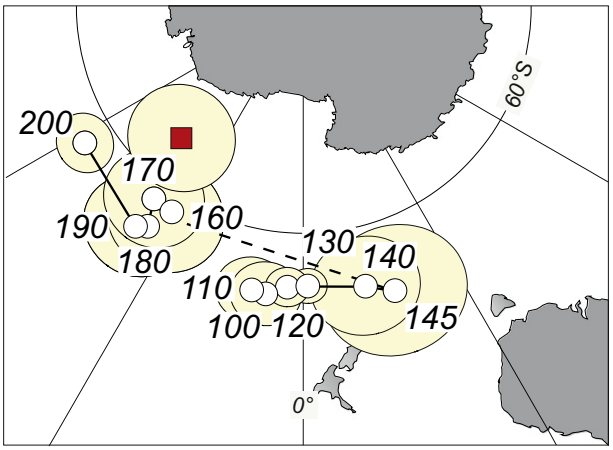


Fig. 6. Apparent polar wander path for Pangea (south poles) after Kent and Irving (2010) with A_{95} confidence ellipses (Fisher, 1953) in East Antarctic coordinates (Lottes & Rowley, 1990). The resulting paleo south pole for this study is shown as a red square. Ages are given in Ma.

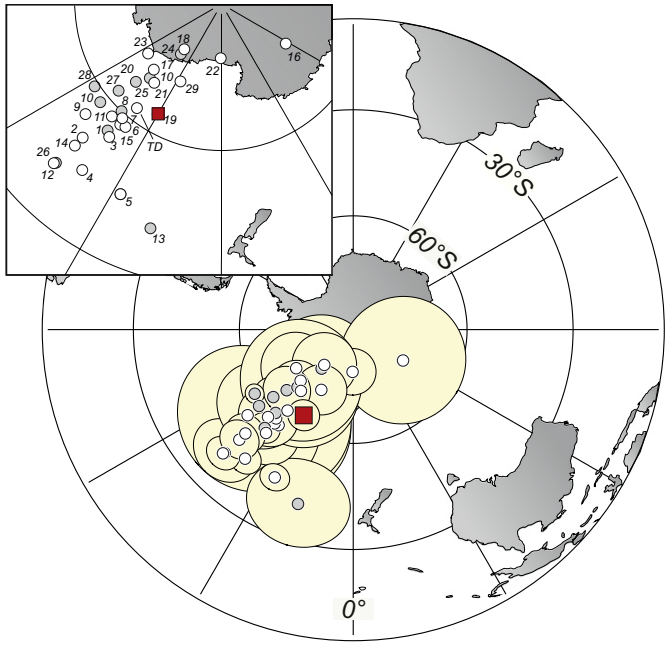


Fig. 7. Paleo South Poles for the Transantarctic Mountains with their associated α_{95} ellipses. Numbers in inset refer to entries in Table 2. Pole positions where secular variations has not been averaged (entries 1,8,10,11,12,20,21,27,28) or are based on less than 16 samples (entries 8, 10, 11, 13, 20, 24) are shaded in light gray. The pole position for the Tasmanian dolerite is marked TD, the resulting mean pole for this study with a red square.

strong resemblance to the data from Vestfjella published by (Løvlie, 1979). The latter fulfill all requirements for high quality data and there is no room for doubt that the reverse magnetizations reported from the basal lavas at Pagodromen and Skansen at Vestfjella are reliable. Based on geochemical data the Vestfjella volcanic rocks of western Dronning Maud Land form part of the Karoo province, however rock ages are still poorly defined geochronologically (Heinonen et al., 2010). It might therefore be tempting to correlate the inversely magnetized basalts from Pagodromen and Skansen with the reverse magnetozone observed in the lower part of the Lesotho basalts (Hargraves et al., 1997) which have been dated using $^{40}Ar/^{39}Ar$ to a range between 179.5 to 186.3 Ma (Duncan et al., 1997). However, single mineral (feldspars) $^{40}Ar/^{39}Ar$ ages for the Vestfjella dyke swarm (Zhang et al., 2003) of 177 ± 1.8 Ma are slightly younger. Clearly, more precise geochronological data is required before a final interpretation can be achieved here. A revised U–Pb age for the Dufek intrusion of 182.7 ± 0.4 Ma (van Groningen et al., 1997) suggest a pre-Ferrar, probably syn-Karoo, age of emplacement for these rocks. Whereas single mineral $^{40}Ar/^{39}Ar$ studies on hornblende yield an age of 179 ± 2 Ma for one silicic dyke. This age defines the closing temperature of the $^{40}Ar/^{39}Ar$ system in hornblende and is rather close to the Curie Temperature of magnetite at 575 °C. The age position of the Mt. Theron is less clear. (Brewer et al., 1996) report $^{40}Ar/^{39}Ar$ ages for 10 dolerites which are in general agreement with other Karoo–Ferrar magmatic rocks (Riley & Knight, 2001) and range from 172.1 ± 1.7 to 181.5 ± 2.5 Ma as well as two dolerites yielding ages of ~190 Ma. In the absence of robust stratigraphic information, the significance of dual polarity paleomagnetic data from Mt. Theron can not be conclusively assessed.

What is obvious from the above is that none of the paleomagnetic studies of volcanic rocks from East Antarctica, including the results presented here, which are dated in a narrow age range around 180 Ma have so far produced magnetization directions of inverse polarity. This supports models of extremely fast emplacement of the Kirkpatrick basalt.

6.4. Spatial distribution of group A and B results

In their review on the TAM, Faure and Mensing (2010) reanalyze the data compilation by Lanza and Zanella (1993) and group those seven VGPs with a latitude between 51°S and 57°S as primary Jurassic and those seven VGPs with latitudes in the range 65°S to 78°S as secondary e.g. Cretaceous in age, leaving ten of the Lanza and Zanella (1993) compilation unaccounted for. However, based on our updated compilation of paleomagnetic pole positions for East Antarctica this bimodal distribution of VGPs cannot be maintained any longer albeit a smeared distribution of VGPs is observed which is confined between roughly 41°S and 79°S. As far as the sometimes sketchy information in the original publications permits, no distinction can be made between results for Kirkpatrick basalts or Ferrar sills or for low-Fe tholeiites (MFCT) vs. high-Fe tholeiites (SPCT). Closer inspection of published demagnetization diagrams reveals a very similar directional pattern in all data accessible. Independently of the demagnetization method applied, univectorial decay of the intensity of the NRM is observed after removal of a viscous overprint at low demagnetization temperatures or low alternating peak fields. If remagnetization is a viable explanation for polar latitudes exceeding 65°S than the Cretaceous remagnetization event which would have affected the TAM along the margin of the Rennick Graben must have wiped out the original magnetization completely in some rocks without affecting others. This, however, seems to be very unlikely. We also note that with the exception of Lanza and Zanella (1993) and Funaki (1984) no study has been published where both high and intermediate polar latitudes have been reported from the same section. Albeit it looks rather like the presence of either high or intermediate polar latitudes is restricted geographically. This is even true for the only publication in which both groups A and B pole positions have been reported (Lanza & Zanella, 1993). However, with the exception of one section (Skinner Ridge) where group B paleopoles within a cluster of group A results have been observed, the presence of either type A or type B directions is geographically controlled. The distribution of sampling areas yielding either group A (polar latitudes <60°S) or group B (polar latitudes >60°S, Fig. 8) can also be observed on a regional scale. Exclusively group A results have been identified in Dronning Maud Land, in volcanic rocks of Karoo type as well as in the more central part of the TAM (Fig. 8a). Directions of both groups have been identified in Northern Victoria Land, however proceeding beyond 73°S only group B directions

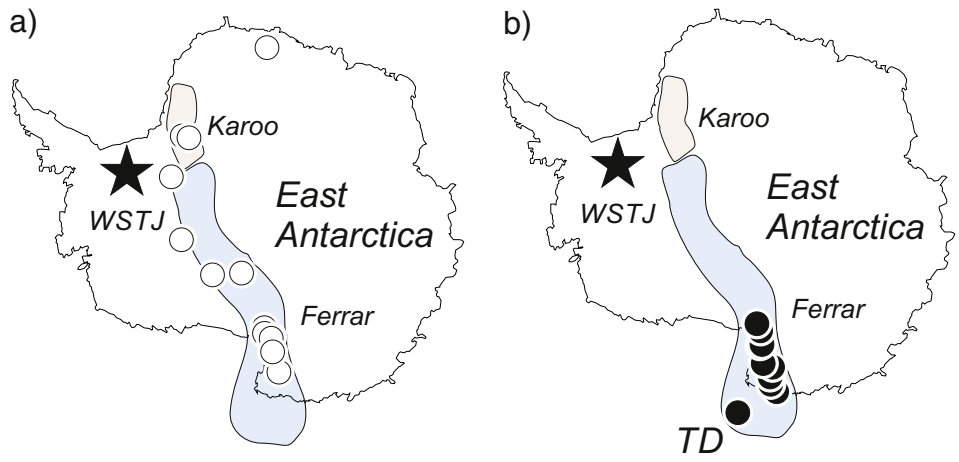


Fig. 8. Sketch map showing the distribution of (a) group A (open circles) and (b) group B locations (full circles). Also shown is the location of the Weddell Sea Triple Junction (WSTJ) and the location of the Tasman dolerites (TD) yielding group B polar latitudes (Irving, 1963) in East Antarctic coordinates (Williams et al., 2011; Whittaker et al., 2013).

are observed (Fig. 8b) suggesting that the chance of identifying group B magnetizations increases from 0 to 100% as a function of distance from the Weddell Sea Triple Junction (WSTJ, Fig. 8).

Antonini et al. (1998) report $^{40}Ar/^{39}Ar$ mineral ages for high-Fe tholeiites of Scarab Peak Type (SCPT, Fleming et al., 1995) which are significantly younger than those obtained for low-Fe tholeiite of Mount Fazio Type (MFT, Fleming et al., 1995) yielding ages of 175.1 ± 1.02 Ma. Unfortunately, however, these results are only available in abstract form and have not been verified up to now. However, Antonini et al. (1997) also demonstrate that there seems to be a relationship between *normal* (group A) and *anomalous* sensu Lanza and Zanella (1993) (group B) pole positions and different styles of magmatic differentiation and that group B magnetizations are linked to MFCT volcanics. The fact that Antonini et al. (1997) were also successful in demonstrating that the volcanic rocks from Northern Victoria Land do not show any Sr-isotopic evidence for a Cretaceous thermal event affecting these rocks.

All this suggests a steady advancement of explosive centers and a continuous younging of the volcanic activity until finally reaching its terminal location in Tasmania (in East Antarctic coordinates). This concept is compatible with models of long distance magma transport (Ferrar sills) as advocated by Leat (2008) and upward migration at a limited number of volcanic centers (Elliot & Fleming, 2004). The occurrence of regionally distinct lava piles either carrying group A or group B magnetizations is not contradicting this model and we argue that the simultaneous presence of group A and group B volcanic rocks in Northern Victoria Land might represent two phases of volcanic activity slightly delayed in time. The streaked-out distribution of paleopole positions, therefore, might reflect different times of NRM acquisition.

7. Conclusion

The rock and paleomagnetic properties of the volcanic rocks from Gair Mesa show very uniform behavior in both their directional and magnetic properties. Both, AF- and thermal demagnetization experiments yield stable and comparable directions in 76% of all samples measured. No significant contribution by a viscous or secondary magnetic overprint has been observed and the resulting site mean directions, although rather steep, do not show resemblance of any post mid Jurassic magnetic field in Antarctica. After removal of a low temperature/low coercivity component of magnetization (component LTCC) which is broadly similar to the direction of the present day magnetic field at the sampling location a high temperature/coercivity component of magnetization (component HTCC) of exclusively normal polarity is present in all samples measured yielding a mean direction of Dec: 285.4° Inc: −78.3° with an α_{95} of 3.4° based on site mean data from 22 flows. No systematic variation in declination/inclination as function of position within the volcanic pile has been observed.

Based on rock magnetic experiments in addition to microscopy imaging, we rule out a remagnetization event as the origin of component HTCC. Therefore we interpret our results as a primary thermomagnetic remanence (TRM) which was acquired during cooling of the basalts. This implies that the whole pile of 40 flows and 800 m were deposited in a rather short time interval, definitively less than 3 Myr taking the longest normal polarity intervals of the relevant polarity timescale into account (Gradstein et al., 2012).

The resulting VGP for the HTCC plots at 66.4°S and 227.7°E on the north-south elongated swath defined by Jurassic paleopoles for East Antarctica (Fig. 7). We note that this swath does not disappear after having removed pole positions, which (a) are of low quality, (b) might have been affected by secular variation, or (c) are based on an insufficient number of samples. Our resultant pole of primary origin is located within group B and therefore disproves the hypothesis that group B poles reflect a remagnetization event of Cretaceous age. We also stress the close similarity of our new pole position and other group B poles to the results from the Tasmanian dolerites (Irving, 1963; Schmidt, 1976) whose reliability is supported by a positive conglomerate test (Irving, 1963).

In accord with the published age dates for the Ferrar dolerites and Kirkpatrick basalts we suggest that both group A and B magnetizations were acquired during emplacement of the magmatic pile within chron e-To-N (Torcian) of normal polarity lasting between 182.5 Ma and 179.5 Ma and only pierced by two rather short episodes of inverse polarity of the Earth magnetic field (Gradstein et al., 2012).

If our line of reasoning is correct and the rocks studied carry (a) a primary magnetization acquired during the late early Jurassic and that alternative interpretations such as a (b) Cretaceous remagnetization or the recording of the (c) onset of a reversal can be ruled out, the only explanation for the distribution of pole positions is a global geomagnetic or geodynamic feature. In a series of recent papers Kent and Irving (2010), Muttoni et al. (2013) have reanalyzed the paleomagnetic database and pointed out the paucity of reliable data for the Jurassic. They have also suggested that the Jurassic segments of recently published apparent polar wander paths (Torsvik et al., 2008, 2012) might be to smooth due to averaging remagnetized data from carbonates (Muttoni et al., 2013) and that the 183 Ma to 151 Ma segment of the Jurassic APWP is ratherundefined (Fig. 6). This has led Muttoni et al. (2013) and Mattei et al. (2014) to postulate massive True Polar Wander during the period in question. Although this is rather speculative, the distribution of pole positions for the TAM presented here

might point to more unknown complexities in the APW for the Jurassic and ultimately support models involving True Polar Wander.

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References

Antonini, P., Antonio, M.D., Civetta, L., Orsi, G., Petrini, R., Piccirillo, E.M., 1997. Geochemical and isotropic signature of Ferrar rocks with different virtual geomagnetic paleopoles Victoria Land, Antarctica. International Symposium on Antarctic Earth Sciences, pp. 505–509.

Antonini, P., Marzolli, A., Demarchi, G., Renne, P., 1998. A late episode of Ferrar magmatism? Geochemistry and preliminary $^{40}Ar/^{39}Ar$ results of high Fe (spct) tholeiites from Southern Victoria Land, Antarctica. Goldschmidt Conference Toulouse.

Beck, M.E., 1972. Palaeomagnetism and magnetic polarity zones in Jurassic Dufek intrusion, Pensecola Mountains, Antarctica. Geophysical Journal of the Royal Astronomical Society 28, 49–63.

Bentley, C., 1991. Configuration and structure of the subglacial crust. The Geology of Antarctica 17, 335–364.

Biggin, A., van Hinsbergen, D., Langereis, C., Straathof, G., Deenen, M., 2008. Geomagnetic secular variation in the Cretaceous Normal Superchron and in the Jurassic. Physics of the Earth and Planetary Interiors 169, 3–19.

Blundell, D., Stephenson, P., 1959. Palaeomagnetism of some doelrite intrusions from the Theron Mountains and Whichaway Nunataks, Antarctica. Nature 184, 1860.

Brewer, T.S., Rex, D., Guise, P.G., Hawkesworth, C.J., 1996. Geochronology of Mesozoic tholeiitic magmatism in Antarctica: implications for the development of the failed Weddell Sea rift system. Geological Society, London, Special Publications 108, 45–61.

Briden, J., Oliver, R., 1963. Palaeomagnetic results from the Beardmore Glacier Region, Antarctica. New Zealand Journal of Geology and Geophysics 6, 388–394.

Bull, C., Irving, E., Willis, I., 1962. Further palaeomagnetic results from South Victoria Land, Antarctica. Geophysical Journal of the Royal Astronomical Society 6, 320–336.

Burmester, R.F., Sheriff, S.D., 1980. Paleomagnetism of the Dufek intrusion, Pensacola Mountains. Antarctical Journal U.S. 15, 43–45.

Cherry, E.M., Noltimier, H.C., 1981. Paleomagnetic Results: Kirkpatrick Basalts At Brimstone Peak And Gorgon Peak, antarctica. p. 555.

Cox, A., 1970. Latitude dependence of the angular dispersion of the geomagnetic field. Geophysical Journal of the Royal Astronomical Society 20, 253–269.

Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and compositional dependence. Physics of the Earth and Planetary Interiors 13, 260–267.

Deenen, M.H.L., Langereis, C.G., van Hinsbergen, D.J.J., Biggin, A.J., 2011. Geomagnetic secular variation and the statistics of palaeomagnetic directions. Geophysical Journal International 186, 509–520.

Delisle, G., Fromm, K., 1984. Paleomagnetic investigation of Ferrar Supergroup rocks of North Victoria Land, Antarctica. Geologisches Jahrbuch B60, 41–55.

Delisle, G., Fromm, K., 1989. Further evidence for a Cretaceous thermal event in North Victoria Land. Geologisches Jahrbuch E38, 143–151.

DiVenere, V.J., Kent, D.V., Dalziel, I.W.D., 1994. Mid-Cretaceous paleomagnetic results from Marie Byrd Land, West Antarctica; a test of post-100 Ma relative motion between East and West Antarctica. Journal of Geophysical Research 99 (15,139-15,139).

DiVenere, V., Kent, D.V., Dalziel, I.W.D., 1995. Early Cretaceous paleomagnetic results from Marie-Byrd-Land, West Antarctica — implications for the Weddellia collage of crustal blocks. Journal of Geophysical Research-Solid Earth 100, 8133–8151.

Duncan, R.A., Hooper, P.R., Rehacek, J., Marsh, J.S., Duncan, A.R., 1997. The timing and duration of the Karoo igneous event, southern Gondwana. Journal of Geophysical Research, Solid Earth 102, 18127–18138.

Dunlop, D., 2002. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc) 2. Application to data for rocks, sediments, and soils. Journal of Geophysical Research 107, 1–15.

Elliot, D.H., 1972. Aspects of Antarctic Geology And Drift Reconstructions. Ohio State University, Institute of Polar Studies.

Elliot, D.H., Fleming, T.H., 2004. Occurrence and dispersal of magmas in the Jurassic Ferrar large igneous province, Antarctica. Gondwana Research 7, 223–237.

Elliot, D.H., Siders, M.A., Haban, M.A., 1986. Jurassic tholeiites in the region of the upper Rennick Glacier, North Victoria Land. In: Stump, E. (Ed.), Geological Investigations In Northern Victoria Land. volume 46 Of Antarctic Research Series. AGU, Washington, DC, pp. 249–265.

Encarnacion, J., Fleming, T.H., Elliot, D.H., Eales, H.V., 1996. Synchronous emplacement of Ferrar and Karoo dolerites and the early breakup of Gondwana. Geology 24, 535–538.

Faure, G., Mensing, T., 1993. K–Ar dates and paleomagnetic evidence for Cretaceous alteration of Mesozoic basaltic lava flows, Mesa Range, Northern Victoria Land, Antarctica. Chemical Geology 109, 305–315.

Faure, G., Mensing, T., 2010. The Transantarctic Mountains: Rocks, Ice, Meteorites And Water. Springer.

Fisher, R., 1953. Dispersion on a sphere. Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences 217, 295–305.

Fleming, T.H., Elliot, D.H., Jones, L.M., Bowman, J.R., Siders, M.A., 1992. Chemical and isotopic variations in an iron-rich lava flow from the Kirkpatrick basalt, North Victoria Land, Antarctica — implications for low-temperature alteration. Contributions to Mineralogy and Petrology 111, 440–457.

Fleming, T.H., Foland, K.A., Elliot, D.H., 1995. Isotopic and chemical constraints on the crustal evolution and source signature of Ferrar magmas, North Victoria Land, Antarctica. Contributions to Mineralogy and Petrology 121, 217–236.

Fleming, T., Heimann, A., Foland, K., Elliot, D., 1997. b. $^{40}Ar/^{39}Ar$ geochronology of Ferrar dolerite sills from the Transantarctic Mountains, Antarctica: implications for the age and origin of the Ferrar magmatic province. Geological Society of America Bulletin 109 (5), 533–546.

Fleming, T.H., Foland, K.A., Elliot, D.H., 1999. Apophyllite $^{40}Ar/^{39}Ar$ and Rb–Sr geochronology: potential utility and application to the timing of secondary mineralization of the Kirkpatrick basalt, Antarctica. Journal of Geophysical Research, Solid Earth 104, 20081–20095.

Funaki, M., 1984. Paleomagnetic investigation of McMurdo sound region, Southern Victoria Land, Antarctica. Memoirs of National Institute of Polar Research, Series C: Earth Sciences 81–81.

Gradstein, F.M., Ogg, J., Schmitz, M., 2012. G.O. A Geological Time Scale 2012 vol. 1. Elsevier, p. 1176.

Grindley, G., Oliver, P.J., 1983. Antarctic earth science. Chapter Palaeomagnetism Of Cretaceous Volcanic Rocks From Marie Byrd Land. Australian Academy of Science, Antarctica, pp. 573–578.

Gunn, B.M., 1962. Differentiation in Ferrar dolerites, Antarctica. New Zealand Journal of Geology and Geophysics 5, 820–863.

Hargraves, R., Rehacek, J., Hooper, P., 1997. Palaeomagnetism of the Karoo igneous rocks in southern Africa. South African Journal of Geology 100, 195–212.

Heimann, A., Fleming, T., Elliot, D., Foland, K., 1994. A short interval of Jurassic continental flood basalt volcanism in Antarctica as demonstrated by $^{40}Ar/^{39}Ar$ geochronology. Earth and Planetary Science Letters 121, 19–41.

Heinonen, J.S., Carlson, R.W., Luttinen, A.V., 2010. Isotopic (Sr, Nd, Pb, and Os) composition of highly magnesian dikes of Vestfjella, western Dronning Maud Land, Antarctica: a key to the origins of the Jurassic Karoo large igneous province? Chemical Geology 277, 227–244.

Hoffman, J., Nairn, A.E.M., Peterson, D.N., 1984. The paleomagnetic investigation of flows and sills from the Queen Alexandra Range, Antarctica. Antarctic Research Series 36 (4), 63–744.

Irving, E., 1963. Paleomagnetism of the Narrabeen chocolate shales and the Tasmanian dolerite. Journal of Geophysical Research 68, 2283–2287.

Irving, E., 1964. Paleomagnetism And Its Application To Geological And Geophysical Problems. Wiley-Interscience, New York.

Irving, E., Parry, L.C., 1963. The magnetism of some Permian rocks from New South Wales. Geophysical Journal of the Royal Astronomical Society 7, 395–411.

Kellogg, K.S., Reynolds, R.L., 1978. Paleomagnetic results from Lassiter Coast, Antarctica, and A test for oroclinal bending of Antarctic peninsula. Journal of Geophysical Research 83, 2293–2299.

Kent, D., Irving, E., 2010. Influence of inclination error in sedimentary rocks on the Triassic and Jurassic apparent pole wander path for North America and implications for Cordilleran tectonics. Journal of Geophysical Research 115, B10103.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society 62, 699–718.

Krása, D., Petersen, K., Petersen, N., 2007. Variable field translation balance. In: Gubbins, D., Herrero-Bervera, E. (Eds.), Encyclopedia Of Geomagnetism And Paleomagnetism. Springer, Dordrecht, The Netherlands, pp. 97–99.

Lanza, R., Zanella, E., 1993. Paleomagnetic of the Ferrar dolerite in the northern Prince Albert Mountains (Victorial Land, Antarctica). Geophysical Journal International 114, 501–511.

Leat, P.T., 2008. On the long-distance transport of Ferrar magmas. Geological Society, London, Special Publications 302, 45–61.

Lottes, A.L., Rowley, D.B., 1990. Reconstruction Of The Laurusian And Gondwanan Segments Of Permian Pangaea vol. 12, pp. 383–395.

Løvlie, R., 1979. Mesozoic palaeomagnetism in Vestfjella, Dronning Maud Land, East Antarctica. Geophysical Journal of the Royal Astronomical Society 59, 529–537.

Luyendyk, B., Cisowski, S., Smith, C., Richard, S., Kimbrough, D., 1996. Paleomagnetic study of the Northern Ford Ranges, Western Marie Byrd Land, West Antarctica: motion between west and east Antarctica. Tectonics 15, 122–141.

Mattei, M., Muttoni, G., Cifelli, F., 2014. A record of the Jurassic massive plate shift from the Garedu formation of central Iran. Geology 42, 555–558.
Matzka, J., Krasa, D., Kunzmann, T., Schult, A., Petersen, N., 2003. Magnetic state of 10–40 Ma old ocean basalts and its implications for natural remanent magnetization. Earth and Planetary Science Letters 206, 541–553.
McElhinny, M., Lock, J., 1996. Iaga paleomagnetic databases with access. Surveys in Geophysics 17, 575–591.
McIntosh, W.C., Kyle, P.R., Sutter, J.F., 1986. Paleomagnetic results from the Kirkpatrick basalt group, Mesa Range, North Victoria Land, Antarctica. Antarctic Research Series 46, 289–303.
McLeod, I., Gregory, C., 1967. Geological investigations along the Antarctic coast between longitudes 108 degrees E and 166 degrees E. Report, 78. Technical Report. Bureau of Mineral Resources, Australia.
Molzahn, M., Wörner, G., Henjes-Kunst, F., Rocholl, A., 1999. Constraints on the Cretaceous thermal event in the Transantarctic Mountains from alteration processes in Ferrar flood basalts. Global and Planetary Change 23, 45–60.
Muttoni, G., Dallanave, E., Channell, J., 2013. The drift history of Adria and Africa from 280 Ma to present, Jurassic true polar wander, and zonal climate control on Tethyan sedimentary facies. Palaeogeography, Palaeoclimatology, Palaeoecology 386, 415–435.
Ostrander, J., 1971. Paleomagnetic investigations of the Queen Alexandra Range, Antarctica. Antarctic Journal of the United States 6 (5), 183–185.
Petersen, N., Eisenach, P., Bleil, U., 1979. Low temperature alteration of the magnetic minerals in ocean floor basalts. Maurice Ewing Series 2, 169–209.
Renne, P.R., Deino, A.L., Walter, R.C., Turrin, B.D., Swisher, C.C., Becker, T.A., Curtis, G.H., Sharp, W.D., Jaouni, A.R., 1994. Intercalibration of astronomical and radioisotopic time. Geology 22, 783–786.
Riley, T.R., Knight, K.B., 2001. Age of pre-break-up Gondwana magmatism. Antarctic Science 13, 99–110.
Riley, T.R., Curtis, M.L., Leat, P., Millar, I.L., 2009. The geochemistry of middle Jurassic dykes associated with the Straumsvola–Tvora alkaline plutons, Dronning Maud Land, Antarctica and their association with the Karoo large igneous province. Mineralogical Magazine 73, 205–226.
Rolf, C., Henjes-Kunst, F., 2003. Paleomagnetic investigations on Palaeozoic and Mesozoic igneous and metasedimentary rocks from Northern Victoria Land, Antarctica. Terra Antartica 10, 157–170.
Rolf, C., Henjes-Kunst, F., 2005. Palaeomagnetic and geochronological study of late Pan-African and Mesozoic igneous and metamorphic rocks from Central Dronning Maud Land, East Antarctica. Geologisches Jahrbuch B97, 7–52.
Schmidt, P.W., 1976. The non-uniqueness of the Australian Mesozoic palaeomagnetic pole position. Geophysical Journal of the Royal Astronomical Society 47, 285–300.
Singer, B.S., Hoffman, K.A., Coe, R.S., Brown, L.L., Jicha, B.R., Pringle, M.S., Chauvin, A., 2005. Structural and temporal requirements for geomagnetic field reversal deduced from lava flows. Nature 434, 633–636.
Stump, E., 1986. Geological investigations in Northern Victoria Land. Volume 46 Of Antarct. Res. Ser. American G.
Stump, E., 1995. The Ross Orogen Of The Transantarctic Mountains. Cambridge University Press, New York (Cambridge University Press, New York).
Tauxe, L., 1998. Paleomagnetic Principles And Practice. Kluwer Academic, Dordrecht.
Tauxe, L., 2010. Essentials Of Paleomagnetism. University of California Press, CaliforniaTauc.
Tessensohn, F., amd Duphorn, K., Jordon, H., Kleinschmidt, G., Skinner, D., Vetter, U., Wright, T., Wyborn, D., 1981. Geological comparison of basement units in North Victoria Land, Antarctica. Geologisches Jahrbuch B41, 31–88.
Torsvik, T.H., Müller, R.D., Van der Voo, R., Steinberger, B., Gaina, C., 2008. Global plate motion frames: toward a unified model. Reviews of Geophysics 46, 1–44.
Torsvik, T.H., der Voo, R.V., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P.V., van Hinsbergen, D.J., Domeier, M., Gaina, C., Tohver, E., Meert, J.G., McCausland, P.J., Cocks, L.R.M., 2012. Phanerozoic polar wander, palaeogeography and dynamics. Earth-Science Reviews 114, 325–368.
Turnbull, G., 1959. Some palaeomagnetic measurements in Antarctica. Arctic 12.
Valet, J.P., Fournier, A., Courtillot, V., Herrero-Bervera, E., 2012. Dynamical similarity of geomagnetic field reversals. Nature 490, 89–94.
Van der Voo, R., 1993. Paleomagnetism Of The Atlantic, Tethys And Iapetus Oceans. Cambridge University Press, Cambridge.
van Groningen, J., Egmond, M., Bloemers, H., Swart, G., Minor, D., Mukasa, S., 1997. Zircon U–Pb and hornblende $^{40}Ar/^{39}Ar$ ages for the Dufek layered mafic intrusion, Antarctica: implications for the age of the Ferrar large igneous province. Geochimica et Cosmochimica Acta 61, 2497–2504.
Wack, M., Gilder, S., 2012. The SushiBar: an automated system for paleomagnetic investigations. Geochemistry, Geophysics, Geosystems 13, 1–24.
Whittaker, J.M., Williams, S.E., Müller, R.D., 2013. Revised tectonic evolution of the Eastern Indian Ocean. Geochemistry, Geophysics, Geosystems 14, 1891–1909.
Williams, S.E., Whittaker, J.M., Müller, R.D., 2011. Full-fit, palinspastic reconstruction of the conjugate Australian–Antarctic margins. Tectonics 30, 1–21.
Zhang, X., Luttinen, A.V., Elliot, D.H., Larsson, K., Foland, K.A., 2003. Early stages of Gondwana breakup: the $^{40}Ar/^{39}Ar$ geochronology of Jurassic basaltic rocks from Western Dronning Maud Land, Antarctica, and implications for the timing of magmatic and hydrothermal events. Journal of Geophysical Research, Solid Earth 108, 1–19.
Zijderveld, J.D.A., 1967. A.C. Demagnetization Of Rocks: Analysis of Results. Elservier, Amsterdam, pp. 254–286.